



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2014

BIOLOGY

9648/01

HIGHER 2

19 September 2014

Paper 1 Multiple Choice

1 hour 15 mins

Additional Material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, pencil clips, highlighters, glue or correction fluid.

Write your name, centre number and index number on the Answer Sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

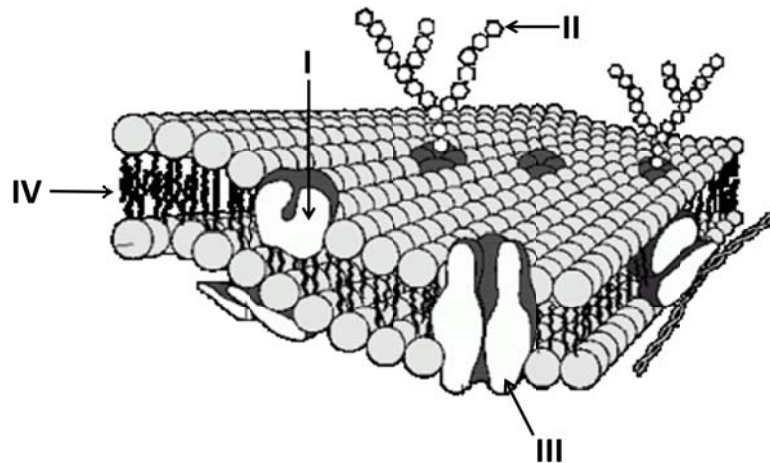
Calculators may be used.

This question paper consists of **21** printed pages.

- 1 A particular genetic disorder causes the progressive degeneration of nerve cells, eventually leading to death. It is caused by a defective enzyme associated with one of the cell's organelles. The enzyme is normally responsible for the breakdown of glycolipids in the cell surface membrane. In the absence of the functional enzyme, accumulation of lipids within the nerve cells leads to the death of these cells.

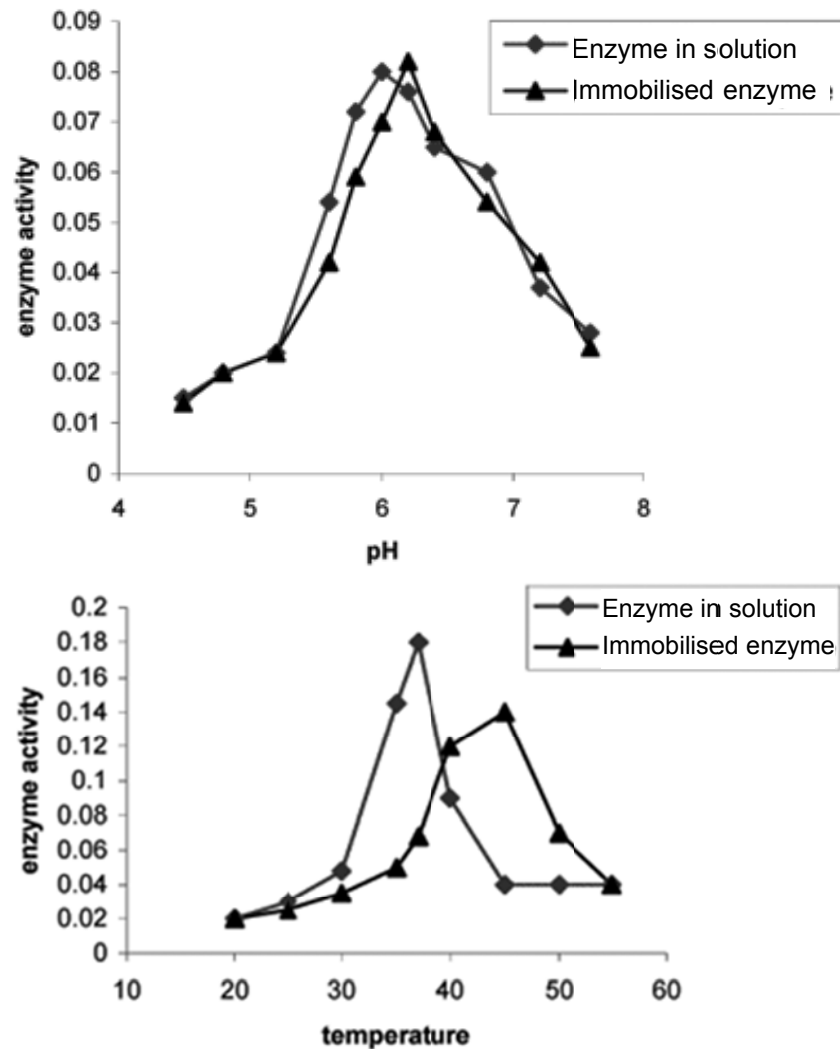
Which organelle is the defective enzyme likely to be associated with?

- A Nucleus
 B Lysosome
 C Rough endoplasmic reticulum
 D Smooth endoplasmic reticulum
- 2 The diagram below shows a section of the cell surface membrane from an arctic fish. Which of the following options regarding the labelled components of the membrane is correct?



	Labelled Cell Component			
	I	II	III	IV
A	Plays a role in cell-cell adhesion	Plays a role in cell-cell recognition	Allows transport of proteins like insulin	Contains a high amount of saturated lipids
B	Plays an enzymatic role	Variation in branching of oligosaccharide allows for cell-cell recognition	Allows transport of amino acids	Contains a high amount of cholesterol
C	Maintains the fluidity of the cell surface membrane	Variation in branching of amino acids allows cell-cell recognition	Plays a role in maintaining membrane potential	Is produced at the Rough Endoplasmic Reticulum
D	Is involved in active transport of ions	Is termed collectively with other glycolipids as the glycocalyx	Is localised to a specific region in the cell surface membrane	Plays a role in cell-cell recognition

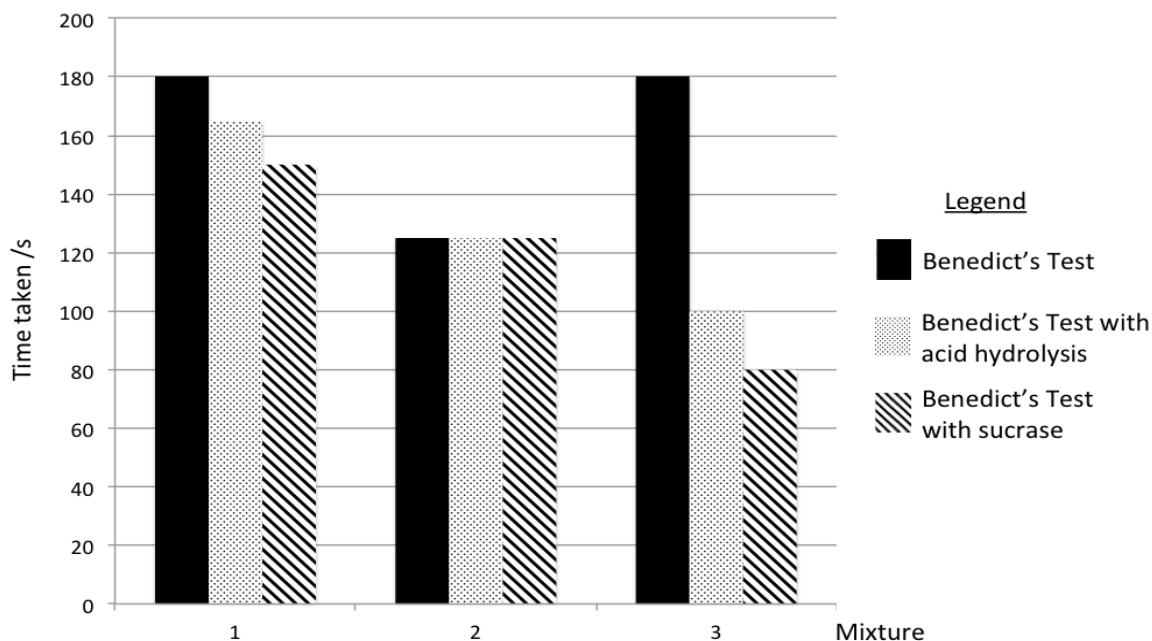
- 3 When used industrially, enzymes are often immobilised on a solid support so that they can be easily removed from the reaction mixture and reused. The following graphs compare the activity of an enzyme in solution against the same enzyme immobilised on a solid support, under different pH and temperature.



Which of the following statements is true?

- A Enzyme activity is about the same at low pH for both the immobilised enzyme and the enzyme in solution but the optimum pH is significantly higher for the immobilised enzyme as it is more resistant to pH changes.
- B At 20°C, the 3D configuration of both the immobilised enzyme and the enzyme in solution is altered, thus accounting for the low enzyme activity.
- C The bonds in the immobilised enzyme remain intact at a higher temperature than the enzyme in solution, thus accounting for the higher enzyme activity above 40°C.
- D The immobilised enzyme gives a higher $V_{\max}$ value under both optimal pH and temperature than the enzyme in solution.

- 4 Three mixtures (1, 2 and 3) containing different concentrations of glucose and sucrose were given to a student. In order to determine the relative concentration of each substance in the mixture, the student subjected the mixtures to different conditions before carrying out Benedict's Test. The time taken for brick-red precipitate to form was recorded in the graphs below.



Which of the following statement can be concluded from these graphs?

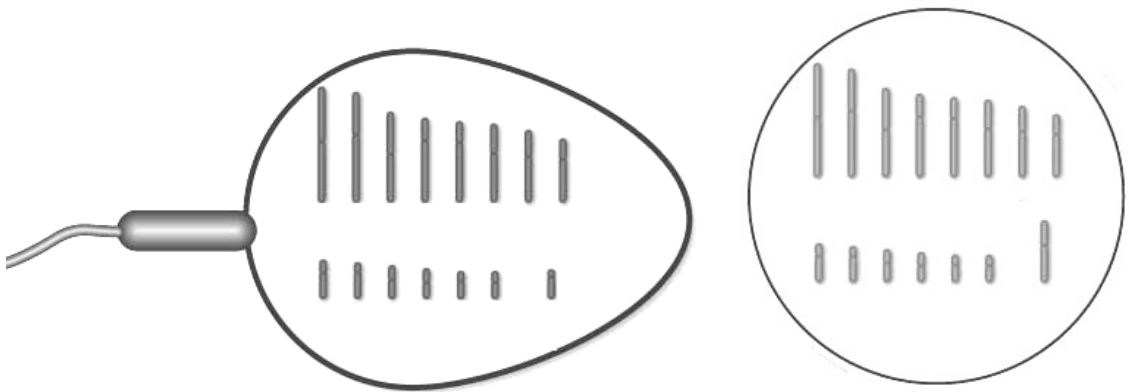
- A Mixture 2 contains equal concentrations of glucose and sucrose.
 B Mixture 1 has a higher concentration of glucose than Mixture 2.
 C Mixture 3 has a higher concentration of sucrose than Mixture 1.
 D Mixture 3 has a lower concentration of sucrose than Mixture 1.
- 5 To identify an unknown substance X, a scientist conducted the following series of tests.

Mixture tested	Test	Result
X mixed with glycogen	Iodine	Red-brown colour
X mixed with starch	Benedict's solution	Orange precipitate
X only	Biuret solution	Violet colour
X only	Ethanol emulsion	Clear

What is substance X likely to be?

- A Maltose
 B Amylose
 C Amylase
 D Fatty acid

- 6 Each of the following statements describes a similarity or difference between mitosis and meiosis. Which statement is false?
- A Mitosis always produces diploid cells while meiosis always produces haploid cells.
 - B Mitosis always results in two daughter cells while meiosis always results in four daughter cells.
 - C Both mitosis and meiosis can occur in the absence of centrioles.
 - D Both mitosis and meiosis involve the separation of sister chromatids.
- 7 The diagram below shows the egg cell and sperm cell formed by one mammal species, as well as the number of chromosomes they contain.



How many DNA molecules would be found in a cell from this organism at prophase II of meiosis II?

- A 15
 - B 30
 - C 60
 - D 120
- 8 Gene mutations involve changes in the nucleotide sequence of DNA. Which of the following descriptions regarding gene mutations is correct?
- A Frame-shift mutations can be caused by base substitution.
 - B Missense mutations can be selectively neutral.
 - C Silent mutations may not be selectively neutral.
 - D Frame-shift mutations can result from an inversion of bases.

9 The mechanism of action of four drugs that inhibit DNA replication is stated below.

- 1 Aphidicholine inhibits DNA polymerase.
- 2 Cytarabine is converted into a molecule that can substitute for a DNA nucleotide and also inhibits DNA repair mechanisms.
- 3 Epirubicin inhibits an enzyme involved in the unwinding and separation of DNA strands.
- 4 Hydroxycarbamide inhibits an enzyme involved in the production of deoxyribonucleotides.

Which row correctly matches the effects of these drugs on DNA replication?

	Effects of Drug on DNA Replication			
	Decreased pool of available nucleotides inhibits chain elongation	DNA damaged during replication and cell death occurs	DNA strands not available as templates for replication	Exposed DNA template strands unable to be copied
A	aphidicholine	cytarabine	epirubicin	hydroxycarbamide
B	hydroxycarbamide	epirubicin	aphidicholine	cytarabine
C	epirubicin	hydroxycarbamide	cytarabine	aphidicholine
D	hydroxycarbamide	cytarabine	epirubicin	aphidicholine

10 An insertion mutation occurs in the gene coding for an enzyme, tyrosinase. Nucleotide sequences of the gene (the non-template strand), as well as the corresponding amino acid sequence of tyrosinase, are shown below.

Wild-type allele	ATG	AAG	TTG	GCT	AAA	TGG	GGA
Wild-type protein	Met	Lys	Leu	Ala	Lys	Trp	Gly
Mutant allele	ATG	AAG	TTA	GGC	TAA	ATG	GGG
Mutant protein	Met	Lys	Leu	Gly	-		



Insertion of adenine

Which feature of the genetic code **cannot** be observed based on the information given?

- A The genetic code is degenerate.
- B The genetic code is punctuated.
- C The code is non-overlapping.
- D The code is universal.

- 11** The enzyme DNase is added to samples taken from the same chromosomal region from three different cell types. After 20 minutes, the remaining DNA samples are weighed and the following results are obtained.

Cell type	Mass of DNA before adding DNase/ μg	Mass of DNA after adding DNase/ μg
P	5.0	5.0
Q	5.0	3.4
R	5.0	4.1

Which of the following statements correctly suggests about the genes and their expressions in the three different cell types?

- A** Some regions of the cell type **P** chromosomes contain DNA coding for rRNA.
 - B** The genes found in the same chromosomal region in the three cell types are expressed the most in cell type **Q**.
 - C** The mass of DNA in cell type **P** did not decrease because the histone tails are acetylated.
 - D** There are more phosphodiester bonds in the DNA of cell type **Q** than those in cell types **P** and **R**.
- 12** The following statements are about eukaryotic control elements:
- 1 Attachment of RNA polymerase II at the TATA box is achieved with the help of a series of specific transcription factors.
 - 2 The DNA binding site on general transcription factors and specific transcription factors is the same.
 - 3 When the histones found in part of a chromosome are methylated, the control elements of a gene are easily accessed.
 - 4 Repressors bind to regions of DNA far upstream of promoters to repress transcription.

Which of the above statement(s) is/ are true?

- A** 1 and 3
 - B** 1 and 4
 - C** 2 and 4
 - D** 4 only
- 13** Telomeres in most cells are shortened with every cell division due to the end replication problem. Which of the following statements explains this?
- A** The telomerase gene is not found in all cells.
 - B** The enzyme telomerase is not activated in most cells.
 - C** The telomerase gene is not expressed in most cells to add a repeated sequence to the 3' end of DNA.
 - D** The telomerase gene is not expressed in most cells to add a repeated sequence to the 5' end of DNA.

- 14 Four different genes are regulated in the following different ways:

Gene **W** is considered a split gene.

Gene **X** undergoes tissue-specific patterns of alternative splicing.

Gene **Y** is made up of mostly coding regions.

Gene **Z** is expressed only in the presence a metabolite.

Which row of the table correctly identifies which genes are prokaryotic and eukaryotic?

	Prokaryotic	Eukaryotic
A	W and Y	X and Z
B	Y and Z	W and X
C	X and Y	W and Z
D	X and Z	W and Y

- 15 Which of the following statements are **not** features of cancer cells undergoing metastasis?

- 1 Presence of cell-to-cell adhesion and loss of telomerase activity
- 2 Lack of cell-to-cell adhesion and presence of high telomerase activity
- 3 Presence of density-dependent inhibition and able to undergo apoptosis
- 4 Lack of density-dependent inhibition and anchorage dependence

- A 1 and 4
B 1 and 3
C 2 and 3
D 2 and 4

- 16 Which of the following pairs of statements is true of generalised and specialised transduction?

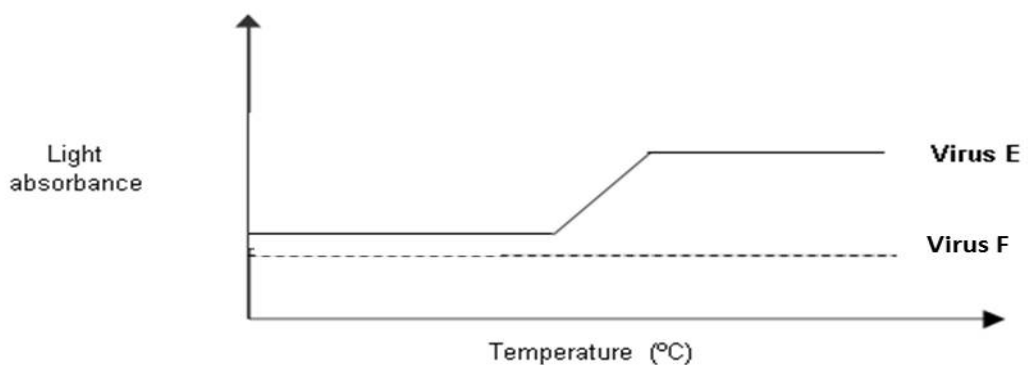
	Generalised transduction	Specialised transduction
A	Viral genome is not transcribed.	Viral genome is transcribed.
B	Viral DNA is replicated by host DNA polymerase.	Viral DNA is replicated by viral DNA polymerase.
C	Viral capsid contains a hybrid chromosome.	Viral capsid contains only bacterial chromosome.
D	Any piece of bacterial DNA can be transferred.	Only a specific set of bacterial genes is transferred.

- 17 A single mutation has occurred in an *E. coli* cell. If the structural genes in a *trp* operon were not expressed regardless of the presence or absence of the amino acid tryptophan, which of the following mutations could have taken place?

- 1 The cell had a mutation at the operator of the *trp* operon.
- 2 The cell had a mutation at the *trpR* gene.
- 3 The cell had a mutation at the promoter of the *trp* structural genes.

- A 3 only
 B 1 and 2
 C 2 and 3
 D 1, 2 and 3

- 18 Two new viruses **E** and **F**, which infect eukaryotic cells, have been identified. In one experiment, the nucleic acid from each virus is isolated and analyzed over a range of temperatures. The light absorbance of nucleic acids changes when denaturation or annealing occurs. The following graph shows the effect of temperature on the light absorbance of the nucleic acids found in viruses **E** and **F**.



In a second experiment, it is found that treatment with RNA-dependent RNA polymerase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus **F** to infect cells. In contrast, RNA-dependent RNA polymerase inhibitors have no effect on the ability of virus **E** to infect cells but DNA synthesis inhibitors block infection by virus **E**.

Which of the following conclusions can be drawn from both the experiments?

- A The genome of virus **E** is single-stranded RNA and that of virus **F** is double-stranded DNA.
 B The genome of virus **E** is double-stranded DNA and that of virus **F** is single-stranded RNA.
 C The genome of virus **E** is double-stranded RNA and that of virus **F** is single-stranded DNA.
 D The genome of virus **E** is double-stranded DNA and that of virus **F** is double-stranded RNA.

19 All of the following statements about viruses are true **except**:

- A The genome of RNA viruses are more likely to mutate than those of DNA viruses.
- B All viruses produce RNA as an intermediate molecule during the production of new viruses.
- C All RNA viruses produce DNA as an intermediate molecule during the production of new RNA viruses.
- D Before entering a host cell, specific proteins of viruses bind to receptors on specific host cells.

20 A couple has three children, all of whom have brown eyes and blond hair. Both parents are heterozygous for brown eyes but one is a blond (rr) and the other is a redhead (Rr).

What is the probability that their next daughter will be a brown-eyed redhead?

- A 1/2
- B 3/8
- C 1/16
- D 3/16

21 Purple buds of the morning glory flower, *Ipomoea*, open into blue flowers. As the flower opens, the pH on the vacuoles of the flower epidermal cells increases and this results in a change of colour from purple to blue.

A mutant purple-flowered morning glory plant carries recessive alleles of a gene **B/b**, coding for a membrane-bound ion pump, and is unable to increase the pH of the vacuole.

Both normal blue flowers and mutant purple flowers have the same anthocyanin pigment, coded by the dominant allele of the gene **A/a**. Plants with **aa** cannot produce anthocyanin and they have white flowers.

The genes **A/a** and **B/b** are not linked.

A blue-flowered morning glory plant was crossed with a purple-flowered plant. Their offspring consisted of plants which are blue-flowered, purple-flowered as well as white-flowered.

What were the genotypes of the blue-flowered and purple-flowered plants?

	Blue-flowered plant	Purple-flowered plant
A	AABB	AaBb
B	AaBb	Aabb
C	AaBB	Aabb
D	AABb	aabb

22 Which of the following statements is true?

- A Continuous variation shows a normal distribution and is only influenced by genetic factors.
- B Continuous variation is controlled by many genes and the traits are usually well defined with no gradation.
- C Discontinuous variation shows traits that follow discrete distribution and is mostly influenced by environmental factors.
- D Discontinuous variation shows traits that are controlled by one or two genes and is relatively unaffected by environmental factors.

23 In an experiment, the farmer crossed pure-bred long grained rice with brown husk and pure-bred short grained rice with white husk. All F_1 offspring showed long grained rice with brown husk. A test cross was made between the F_1 plant and a pure-bred plant with short grained rice with white husk.

The phenotypes of 200 offspring of the test cross was recorded and shown in the table below.

Long grained rice with brown husk	Short grained rice with brown husk	Long grained rice with white husk	Short grained rice with white husk
84	12	15	89

A chi-square test was carried out to determine if the results of the test cross depart significantly from the expected ratio when the genes for rice length and husk colour segregate independently.

degrees of freedom	Distribution of χ^2				
	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$v = c - 1$$

keys to symbols

Σ - 'sum of...'

O - observed value

E - expected value

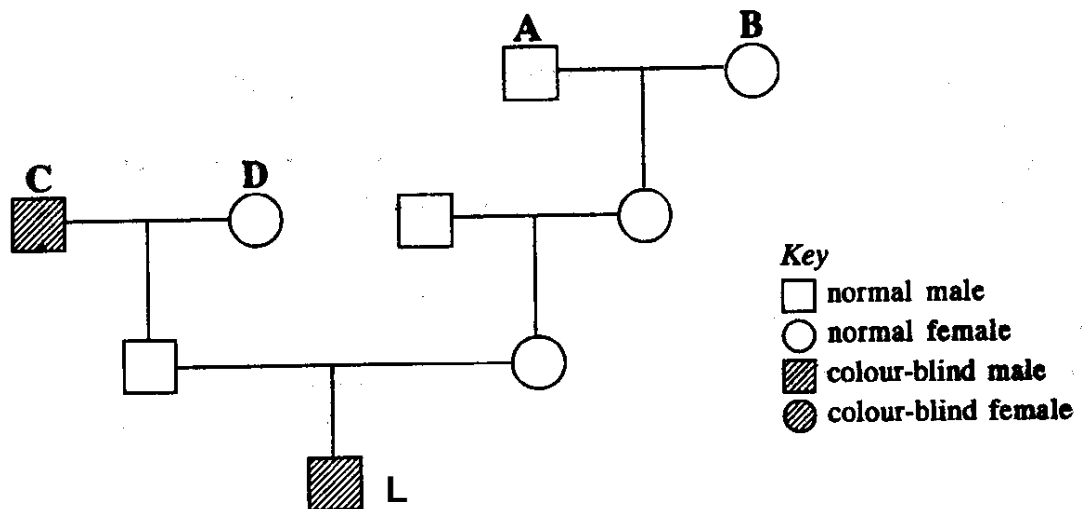
v - degrees of freedom

c - number of classes

Which of the following statements is true?

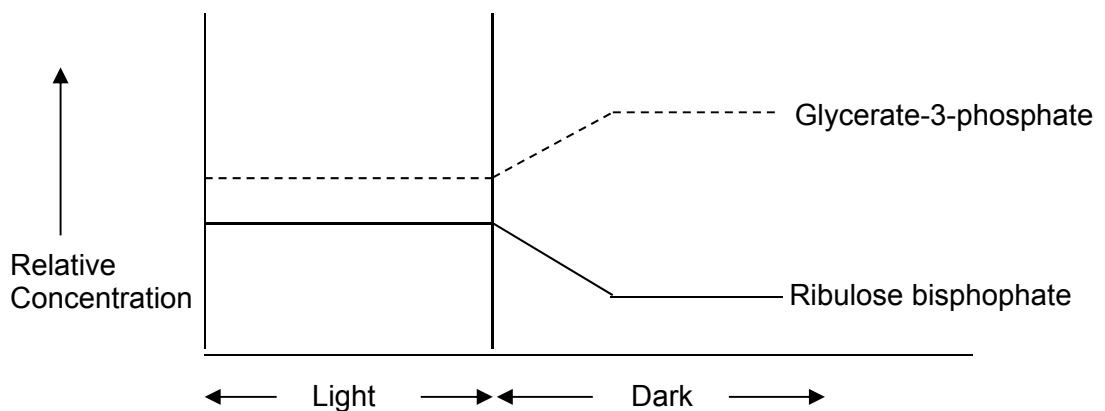
- A The expected ratio of the test cross is 9:3:3:1.
- B The χ^2 calculated is 148.3.
- C At 3 degree of freedom, χ^2 falls at $p < 0.001$.
- D $p < 0.05$, therefore, there is no significant differences between the observed and expected values, therefore the ratio is not 1:1:1:1.

- 24 The diagram shows the pedigree of a family carrying the sex-linked allele for colour-blindness.



From which labelled member of his family, A, B, C or D, did L inherit this disorder?

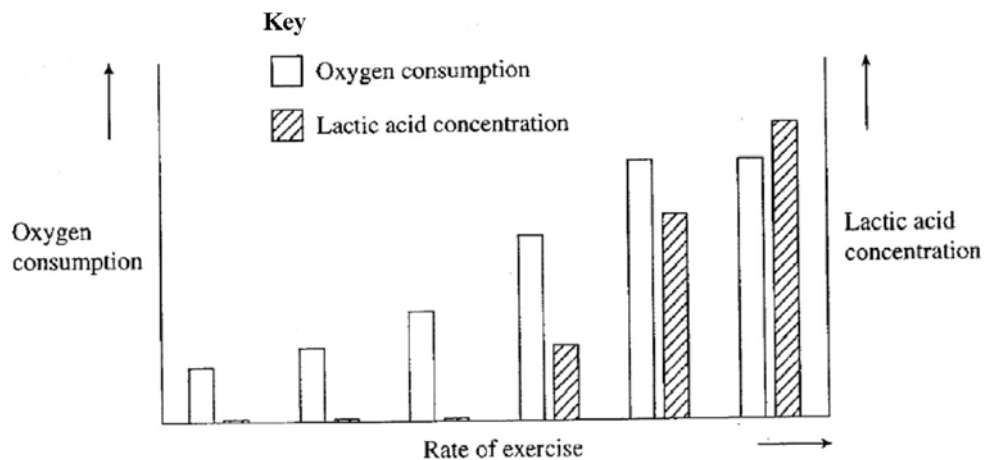
- 25 The graph below shows the effects of a change from light to dark conditions on the relative concentrations of glycerate-3-phosphate (GP) and ribulose biphosphate (RuBP) in a plant.



Which of the following features of the graph does **not** support the hypothesis that RuBP is converted to GP? The compounds

- A have constant concentrations at the start of the experiment.
- B respond immediately to the changed condition.
- C achieve a new steady concentration at the same time.
- D have the same rate of response, one negative and the other positive.

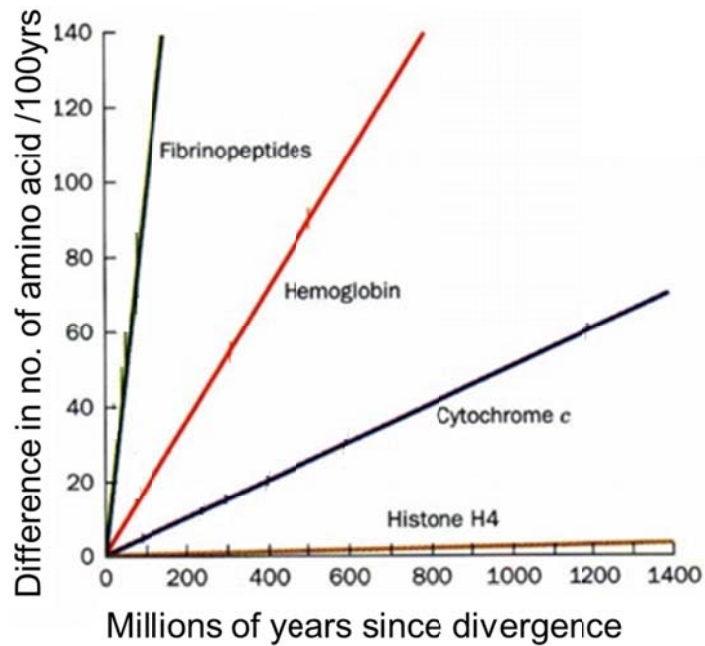
- 26 Scientists measured the oxygen consumption of an athlete at different rates of exercise. They also measured the lactic acid concentration in the athlete's muscles. The results are shown in the graph below.



Which of the following can be deduced from the graph?

- A Anaerobic respiration occurs at a constant rate.
 B At a higher rate of exercise, rate of anaerobic respiration in muscle cells increases.
 C Oxygen consumption increases proportionally with increased rate of exercise.
 D Increase in lactic acid concentration is due to increased oxygen consumption.
- 27 A growth hormone gene from the Chinook salmon was injected into the late embryo stage of the Atlantic Salmon. The resulting Atlantic Salmon is a mosaic salmon with some of its cells containing the gene for the growth hormone.
- In an ecology experiment, five of such genetically-modified (GM) Atlantic salmon were released into rivers in North America, the natural habitat of Atlantic salmon.
- Based on the information given, which of the following statements are explanations on the possible outcome of the Atlantic salmon population in these rivers?
- 1 The GM Atlantic Salmon may flourish as it has a faster growth rate than the wild type, hence it would have a higher reproductive success and be selected for by natural selection.
 - 2 The GM Atlantic Salmon may not flourish in the population as effects of genetic drift would have a greater influence than natural selection.
 - 3 The GM Atlantic Salmon may not flourish as the cells of its gonads may not contain the gene for Chinook's growth hormone.
 - 4 The GM Atlantic Salmon may flourish in an environment where food and mating resources are sparse.
- A 1 and 4
 B 2 and 3
 C 1, 2 and 3
 D All of the above

- 28 The figure below shows the rate of change of amino acids in 4 proteins found in species X and Y over a period of time.

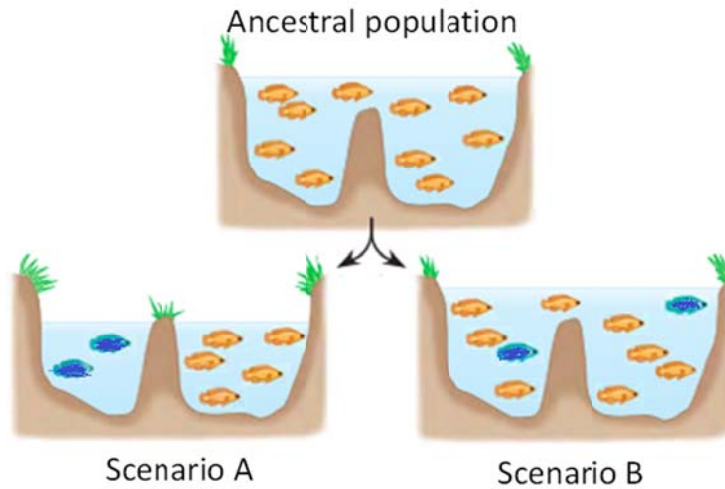


Which of the following statements are correct deductions made from the graph?

- 1 Mutations that resulted in the differences in amino acid in X and Y may be deleterious or beneficial.
- 2 Mutations that resulted in the differences in amino acid in X and Y is neutral.
- 3 Mutations that resulted in the differences in amino acid in X and Y is fixed by natural selection.
- 4 Fibrinopeptide is a better molecule to measure the time of divergence between ape and human than that between chicken and human.

- A** 1 and 2
B 2 and 4
C 3 and 4
D 1 and 3

- 29 The diagram below shows two scenarios **A** and **B** of two different species of fish originating from the ancestral population.

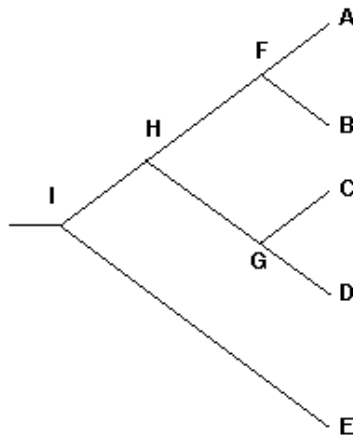


Which of the following statements are true about the two scenarios?

- 1 There is gene flow between the two species of fishes in scenario **B** to produce viable and fertile offspring but not in **A**.
- 2 Preferential mating occurs in scenario **B** to give rise to two separate species but not in **A**.
- 3 Disruptive selection occurs in scenario **B** while directional selection occurs in scenario **A** to give rise to two separate species.
- 4 There is reproductive isolation in scenario **A** but not in **B**.

- A** 1 and 2
B 2 and 3
C 1 and 4
D 2, 3 and 4

- 30 The diagram below shows a phylogenetic tree of different organisms **A**, **B**, **C**, **D** and **E**.



Which of the following conclusion(s) can be made about the relationships between the different organisms?

- 1 Organisms **A**, **B**, **C** and **D** share more common characteristics than that with organism **E**.
- 2 Organisms **A** and **B** share more common characteristics than that between **C** and **D**.
- 3 Organisms **A** and **B** belong to one genus while organisms **C** and **D** belongs to another genus.
- 4 A hierarchical relationship can be seen where **H** represents the class where both orders **F** and **G** belong to.

- A** 1 only
B 1, 2 and 3
C 3 and 4
D 1 and 4

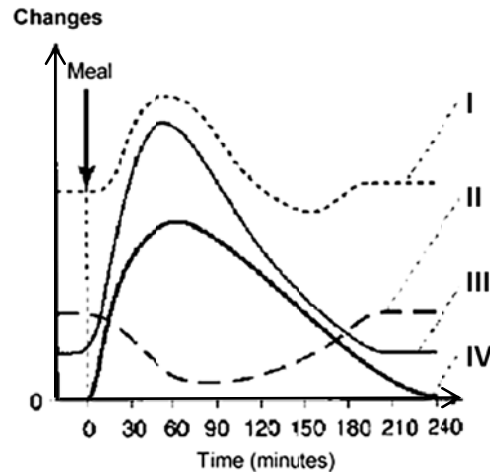
- 31 The table below shows several drugs and their effects on the postsynaptic membrane.

Drug	Effect
1	Induces the opening of chemically gated Na^+ ion channels
2	Induces the opening of voltage-gated Na^+ ion channels
3	Inhibits the transfer of ions through the Na^+/K^+ pump
4	Induces the opening of chemically gated K^+ ion channels

Which drug(s) would increase the chances of generating an inhibitory postsynaptic potential at the junction of an inhibitory synapse?

- A** 4 only
B 2 and 3
C 3 and 4
D 1, 2 and 4

- 32 The figure below shows changes in the levels of insulin, blood glucose, gluconeogenesis, and glucose taken up by muscles cells over time after a meal.



Which of the following options correctly identifies and explains the levels of insulin, blood glucose, gluconeogenesis and glucose uptake by muscles cells to their corresponding graphs?

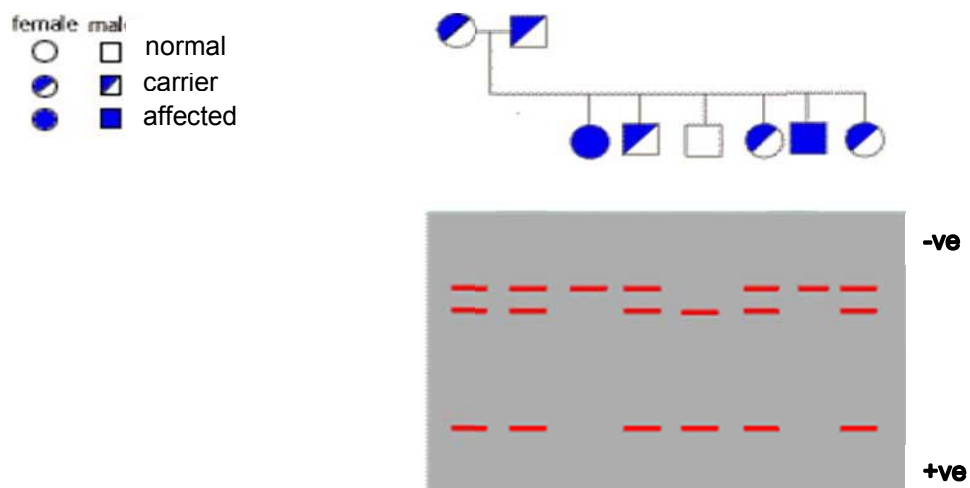
	I	II	III	IV
A	<u>Insulin level</u> Increases due to increased glucose binding to receptors in the β cells	<u>Glucose uptake by muscles cells</u> Increases after meal to decrease blood glucose level	<u>Blood glucose level</u> Is maintained at a constant level due to negative feedback	<u>Rate of gluconeogenesis</u> Decreases due to effects of insulin on insulin receptors in liver cells
B	<u>Rate of gluconeogenesis</u> Decreases due to effects of insulin on insulin receptors in liver cells	<u>Insulin level</u> Increases due to increased glucose binding to receptors in the β cells	<u>Glucose uptake by muscles cells</u> Increases after meal to decrease blood glucose level	<u>Blood glucose level</u> Is maintained at a constant level due to negative feedback
C	<u>Blood glucose level</u> Is maintained at a constant level due to negative feedback	<u>Rate of gluconeogenesis</u> Decreases due to effects of insulin on insulin receptors in liver cells	<u>Insulin level</u> Increases due to increased glucose binding to receptors in the β cells	<u>Glucose uptake by muscles cells</u> Increases after meal to decrease blood glucose level
D	<u>Glucose uptake by muscles cells</u> Increases after meal to decrease blood glucose level	<u>Blood glucose level</u> Is maintained at a constant level due to negative feedback	<u>Rate of gluconeogenesis</u> Decreases due to effects of insulin on insulin receptors in liver cells	<u>Insulin level</u> Increases due to increased glucose binding to receptors in the β cells

- 33** A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T is lipid soluble and diminishes the effect of glucagon. The levels of cAMP were also low in the cell. Which of the following are possible statements that explain the effects of Poison T?

- 1 Poison T binds directly to proteins in the cytoplasm.
- 2 Poison T prevents G protein from hydrolysing GTP.
- 3 Poison T inactivates the enzyme adenyl cyclase.
- 4 Poison T prevents signal amplification by binding competitively to protein kinase A.

- A** 1 and 2
B 3 and 4
C 1, 3 and 4
D 1, 2 and 4

- 34** The diagram below shows an autoradiograph, post-gel electrophoresis, showing the restriction fragments obtained from various members of a family with respect to a disease.

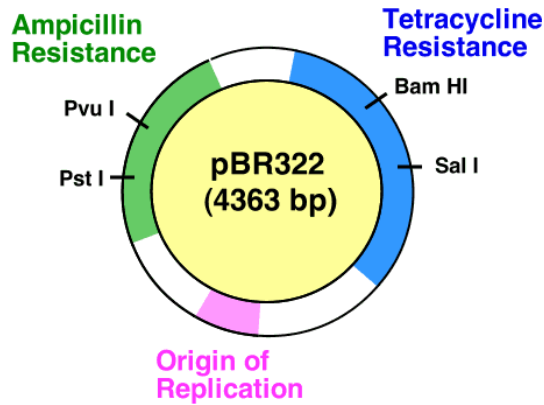


Which of the following statements can be concluded from this autoradiograph?

- 1 The recessive allele is missing a restriction site.
- 2 The disease is autosomal recessive.
- 3 The radioactive probe is able to bind to a specific sequence on the recessive allele only.

- A** 1 and 2
B 2 and 3
C 1 and 3
D All of the above

- 35 The figure below shows an ampicillin-resistant, tetracycline-resistant plasmid, pBR322 (not drawn to scale). In the preparation of a *Drosophila* genomic library, pBR322 is cleaved with *Pst*I before it is ligated with *Pst*I-digested *Drosophila* DNA. The resulting mixture is then used to transform *E. coli*.



Which antibiotic should be added to the medium to screen for recombinant clones and what growth pattern should be selected to obtain recombinant clones containing *Drosophila* DNA inserts?

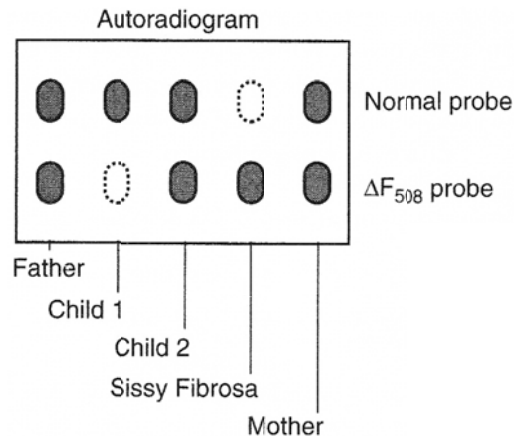
	Antibiotic to be added	Growth pattern for selection
A	Ampicillin	Colonies that grow on ampicillin medium should be tested for growth on a tetracycline medium by replica plating. Colonies that failed to grow with tetracycline probably contain the <i>Drosophila</i> DNA insert.
B	Ampicillin	Colonies that failed to grow on ampicillin medium should be transferred to a medium with X-gal by replica plating. White colonies formed should probably contain the <i>Drosophila</i> DNA insert.
C	Tetracycline	Colonies that grow on tetracycline medium should be tested for growth on an ampicillin medium by replica plating. Colonies that failed to grow on ampicillin probably contain the <i>Drosophila</i> DNA insert.
D	Tetracycline	Colonies that failed to grow on tetracycline medium should be transferred to a medium with X-gal by replica plating. White colonies formed should probably contain the <i>Drosophila</i> DNA insert.

- 36 Which of the following is **not** a benefit of the Human Genome Project?

- A Identification of genes linked with specific genetic disorders.
- B Screening populations for heterozygous carriers.
- C Improving the understanding of evolutionary links.
- D Producing new drugs that target cancer cells in humans.

- 37** A method that uses allele-specific oligonucleotide probes has been developed to detect cystic fibrosis. Oligonucleotide probes, complementary to the region where a 3-base deletion is located, have been chemically synthesised. One probe binds to the mutant (ΔF_{508}) allele whilst the other binds to the normal allele.

DNA from Sissy Fibrosa, her parents and two siblings was isolated and amplified. Their DNA samples were blotted on nitrocellulose paper and treated with oligonucleotide probes. The results obtained were shown in the figure below. Dark spots indicate binding of probe.



Which members of Sissy's family have cystic fibrosis, which are normal and which are carriers?

	cystic fibrosis	normal	carriers
A	Child 1	Sissy Fibrosa	Father, Mother, Child 2
B	Child 1	Father, Mother, Child 2	Sissy Fibrosa
C	Sissy Fibrosa	Father, Mother, Child 2	Child 1
D	Sissy Fibrosa	Child 1	Father, Mother, Child 2

- 38** Gene therapy is the insertion of a therapeutic gene into a target cell with non-functional genes. Which of the following are techniques of gene therapy to treat a genetic disease?

- 1 A therapeutic gene is inserted into a non-specific location within the genome.
- 2 A therapeutic gene is inserted into a specific location within the genome.
- 3 A therapeutic gene enters the cell, but is not inserted into the genome.
- 4 A therapeutic gene is inserted in reverse orientation with respect to the promoter (antisense technology).

- A** 1 and 3
B 2 and 4
C 1, 2 and 3
D 1, 2 and 4

39 Which row shows the normal functions of different types of human stem cells?

	Functions		
A	Cells that can differentiate into bone cells in the skeleton.	Cells that can differentiate into all types of cells during the first 3 days of embryo development.	Cells that can be used for gene therapy to treat ADA-SCID.
B	Cells that can differentiate into cardiac muscle cells in the heart.	Cells that can differentiate into all types of cells during the first 3 days of embryo development.	Cells that can produce more red blood cells to replace worn out cells.
C	Cells that can produce more insulin in damaged pancreas.	Cells that can differentiate into all types of cells in the blastocyst stage of embryo development.	Cells that can differentiate into cartilage cells in the joints.
D	Cells that can differentiate into nerve cells in the brain.	Cells that can differentiate into all types of cells in the blastocyst stage of embryo development.	Haematopoietic stem cells that can produce differentiated cells in the brain.

40 One type of genetically-modified corn has

- a gene for the production of BT toxin which protects the plant against specific insect;
- a '*pat*' gene for tolerance to the herbicide 'Basta' which is used for selection of transformed plants.
- an '*amp*' gene which was introduced in the plant together with the BT toxin gene.

Which of the following is **not** a valid concern regarding mass propagation of such genetically modified corn?

- A** The consumption of the toxin produced by the genetically-modified corn may have negative physiological effects in human.
- B** The consumption of new proteins produced by the genetically-modified corn may cause allergic reactions in human.
- C** The '*amp*' gene may be transferred to the cells lining the intestine tract making treatment with ampicillin ineffective for disease caused by bacteria.
- D** Gene transfer by cross breeding with wild plants may result in the emergence of super weeds.

End of Paper